10 September 1973
SUPERSEDING
MIL-M-55220(EL)
14 March 1963

MILITARY SPECIFICATION
MAST AB-577( )/GRC

This specification is approved for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 This specification covers one type of uniformly sectioned, portable, guyed mast, which can be quickly erected to heights up to 50 feet by two men; designated Mast AB-577( )/GRC. (See 6.1 and 6.5)

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

O-L-164	Leather Dressing, Mildew-preventive
QQ-S-781	Strapping, Steel, Flat and Seals
CCC-C-419	Cloth, Duck, Cotton, Unbleached, Plied-yarns, Army and Numbered
CCC-D-950	Dyeing and Aftertreating Processes for Cotton Cloths
PPP-B-585	Box, Wood, Wirebound
PPP-F-320	Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade), and cut Shapes

MILITARY

MIL-P-116	Preservation: Methods of
MIL-B-117	Bag, Sleeve and Tubing-Interior Packaging
MIL-C-11133	Crate, Wood, Open, Wirebound
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Signal Equipment

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STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-252	Wired Equipment. Classification of Visual and Mechanical Defects
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-731	Quality of Wood Members for Containers and Pallets
MIL-STD-810	Environmental Test Methods

DRAWINGS

ELECTRONICS COMMAND

SC-DL-423549	Mast AB-577( )/GRC
SM-C-423619	Roll Assembly

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 Classification of requirements.- The requirements specified herein are classified as follows:

a. First article requirements. (See 3.2)

b. Construction and process requirements (See 3.3)

c. Material requirements. (See 3.4)

d. Performance requirements. (See 3.5)

3.2 First article requirements.- When specified in the contract or purchase order, the contractor shall furnish two first article units. One unit, selected at random, shall be capable of meeting the requirements of the following paragraphs, in the order given: (See 4.3)

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3.3.9 Requirements covered in subsidiary documents.
3.3.8 Workmanship requirements.
3.5.1 Brake locking and holding requirements.
3.5.2 Coupling closing requirement.
3.3.10 Interchangeability requirements.
3.5.3 Torsion transfer requirements.
3.5.4 Joint bending requirement.
3.5.5 Bounce, loose cargo requirement.
3.5.6 Shock, drop requirement.
3.5.7 Static load requirements.
3.4.1 Tropicalization requirement.

The remaining unit shall be capable of meeting the requirements of the following paragraphs, in the order given:

3.3.9 Requirements covered in subsidiary documents.
3.3.8 Workmanship requirements.
3.5.1 Brake locking and holding requirements.
3.5.2 Coupling closing requirement.
3.3.10 Interchangeability requirements.
3.5.8.1 & 3.5.8.2 Temperature exposure requirements.
3.5.8.3 Humidity exposure requirement.
3.5.8.4 Salt atmosphere exposure requirement.

3.3 Construction and process requirements.-

3.3.1 Construction.- The mast shall be constructed in accordance with the requirements of the drawings listed on SC-DL-423549. (See 3.3.8)

3.3.2 Cleaning.- (See 3.3.8)

3.3.2.1 Parts.- After fabrication, parts shall be cleaned in accordance with requirement 9 of MIL-STD-454 except as specified in an applicable document.

3.3.2.2 Units.- After assembly, units shall be cleaned thoroughly and shall be free from particles of foreign material. In addition, such cleaning shall also be performed before final assembly of the equipment.

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3.3.3 Finish.- The equipment shall be finished in accordance with Specification MIL-F-14072 and the equipment drawings (See 3.3.8 and 3.3.9)

3.3.4 Marking.- Marking shall conform to Specification MIL-M-13231 and the equipment drawings. (See 3.3.8 and 3.3.9)

3.3.5 Riveting.- Riveted joints shall be tight, the joined parts shall be undamaged, and the rivet heads shall be properly seated and tight against their bearing surfaces. (See 3.3.8)

3.3.6 Securing of parts.- Brackets, lugs, flanges, inserts, bolts, and other mounting arrangements shall be suitably tight to retain items securely when subjected to specified tests. (See 3.3.8)

3.3.7 Welding.- Welding shall conform to the equipment drawings and shall be such that, except were called for on the drawings, grinding on the finished weld will be unnecessary. (See 3.3.8)

3.3.7.1 Cleaning prior to welding .- Surfaces to be welded shall be cleaned in accordance with good commercial practice and shall be free from rust, scale, paint, grease, and other foreign material.

3.3.7.2 Process.- Preheating shall be employed where distortion is likely to result from welding. Welds shall have thorough penetration and good fusion and shall be free from scabs, blisters, abnormal pock marks, cracks, voids, slag inclusions, and other harmful defects.

3.3.7.3 Cleaning after welding- Welded assemblies shall be cleaned to remove rust, scale, oxidation products, and excess flux by sandblasting, wire brushing, or other suitable means. Acid used for cleaning shall be completely neutralized and removed.

3.3.8 Workmanship.- The equipment shall be manufactured and assembled in accordance with requirement 9 of MIL-STD-454 and with the following paragraphs herein:

3.3.1 Construction
3.3.2 Cleaning
3.3.3 Finish

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3.3.4 Marking
3.3.5 Riveting
3.3.6 Securing of parts
3.3.7 Welding

3.3.9 Requirements covered in subsidiary documents.- The equipment shall be manufactured and assembled in accordance with the requirements of the following subsidiary documents as covered in the referenced paragraphs herein: (See 4.4)

Document	Paragraph where referenced
MIL-F-14072	3.3.3
MIL-M-13231	3.3.4

3.3.10 Interchangeability.- The following dimensions shall be gaged or measured to insure interchangeability of the mating parts: (See 4.9)

a. Dimensions and location of torsional transfer elements.

b. Mating dimensions of section end fittings and couplings.

c. Mounting dimensions of separable assemblies.

d. Dimensions and location of locating pins for the hand crank.

3.4 Material requirements.-

3.4.1 Tropicalization.-

3.4.1.1 Treatment of materials.- Fungus-inert materials shall be used where practicable. These materials shall be in accordance with requirement 4 of MIL-STD-454. When materials are required to be treated, they shall be cleaned in accordance with requirement 9 of MIL-STD-454 and then shall be treated in accordance with 3.4.1.2 thru 3.4.1.5 and as follows:

a. Linen thread, cotton thread and duck shall conform to requirement 44 of MIL-STD-454, except that, when duck is not required to be fire resistant, it shall conform to CCC-C-419 and shall be given Class B after treatment, using inhibitors "a", "c", "f", or "g" as specified in CCC-D-950. Inhibitor "a" shall be used when the duck will come in contact with natural or synthetic rubber or prolonged contact with skin can occur.

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The color of the fabric shall be olive-drab No. 7. The color of thread shall match shade S-1 of the United States Army Standard Color Card for Official Standardization Shades of Securing Threads.

b. Leather shall be treated with dressing material, type I or II, conforming to O-L-164.

3.4.1.2 Treating compounds.- Treating compounds containing a mercury-bearing fungicide shall not be used. The contractor shall determine that the treating compound is compatible with the material or surface to be treated. Selection of treating compound shall be such that there shall be no increase in flammability of the treated material.

3.4.1.3 Toxicity.- Treating compounds shall cause no skin irritation or other injury to personnel handling the treated material during fabrication, transportation, operation or maintenance of the equipment, or during use of the finished items when used for the purpose intended.

3.4.1.4 Flexibility.- Treatment shall not affect the flexibility of the treated material, to the extent that the equipment may fail to meet specified requirements when subjected to specified service conditions.

3.4.1.5 Statement of treatment.- The statement of treatment shall indicate whether or not nutrient materials have been employed. If nutrient materials have been used, it shall also describe in detail the materials to be treated and the treating compounds and processes that the contractor proposes to use. (See 6.21)

3.4.1.6 Fungus.- The equipment shall be capable of being subjected to Method 508, Procedure I of MIL-STD-810 with no evidence of viable fungus or corrosion. Corrosion is any visible degradation that can be attributed to flaky, pitted, blistered, or otherwise loosened finish or metal surface. The equipment shall be capable of meeting this requirement when those items normally stored in bags are removed and exposed.

3.5 Performance requirements.-

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3.5.1 Brake locking and holding .- The brake, when locked on a mast section, using a locking force not exceeding 30 pounds, shall be capable of withstanding a 500 pound load, applied axially to the mast, without yield or slippage. It shall be capable of meeting this requirement under the following conditions:

a. with the launcher in the vertical position.

b. with the elevator disengaged from the mast section.

c. with the locking force applied at the lock pin hole in the brake handle.

3.5.2 Coupling closing.- The clamps shall be capable of coupling the end fittings of the mast sections with a force no smaller than 2 pounds and no larger than 75 pounds. It shall be capable of meeting this requirement when all eight mast sections are randomly coupled into one continuous length with no selective fitting of joints or clamps.

3.5.3 Torsion transfer.- The eight mast sections, coupled into one continuous length, shall be capable of sustaining a 300 inch pound torque with a resulting angular displacement, over the entire length, or not more than 2 degrees. It shall be capable of meeting this requirement under the following conditions:

a. with the entire assembly freely supported to allow rotation about its axis.

b. with a measurement taken of the torque required to rotate the assembly in the supports.

c. with one end of the assembly then fixed and a torque applied to the other end equal to 300 inch pounds plus one half the torque required to rotate the assembly.

d. with this torque first applied in a clockwise direction and then in a counterclockwise direction, the required angular displacement shall be taken as one half of the total angular displacement between these two extremes.

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3.5.4 Joint bending.- The joints shall be capable of sustaining a 1,600 pound load with no yield or failure of any part of the mast sections. It shall be capable of meeting this requirement under the following conditions:

a. with two mast sections coupled together and simply supported over a span of 100 inches inside to inside of supports.

b. with the 1,600 pound load applied at the coupling.

c. with the coupling positioned centrally in the span.

d. with the supports and load each contacting the mast sections for no more than 120° of circumference and 3 inches of length.

e. after meeting the requirements of 3.5.3

3.5.5 Bounce, loose cargo.- The launcher, with mast sections and carrier prepared for transit, and the other components prepared for transit in the guy assembly and anchor wrap bags, shall be capable of being subjected to Method 514, Procedure XI, Part 2 of MIL-STD-810 and the following conditions with no physical damage except minor surface abrasion and with no degradation in performance:

a. with the tester operated, shafts in phase, at a speed of 250 ± 2 rpm.

b. with the launcher and each bag tested separately or fenced separately.

3.5.6 Shock, drop.- The launcher, with mast sections and carrier prepared for transit, shall be capable of being subjected to Method 516, Procedure II of MIL-STD-810 and the following conditions with no weld cracks, connector failures, deformation of members or other structural damage and with no degradation in performance. Denting of the point of contact is permissible:

a. with the weight taken as over 200 pounds and less than 1,000 pounds and with the largest dimension taken as over 60 inches.

b. with 2 drops performed on each corner, of the side on which winch mounts, for a total of 8 drops.

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3.5.7 Static load.- The completely extended mast, with brake engaged, shall be capable of withstanding the installation and loading conditions of both 3.5.7.1 and 3.5.7.2 with no failure or permanent deformation to any part and no slipping of the guys thru the tensioners. It shall be capable of meeting this requirement after meeting the requirement of 3.5.6.

3.5.7.1 Fully guyed.- The fully guyed mast shall be capable of meeting the requirement of 3.5.7 under the following conditions:

a. with each guy pretensioned to 150 pounds.

b. with the following forces all applied simultaneously on the same side of the mast in a plane which bisects the angle between two adjacent guys:

(1) an 80 pound weight applied axially to the mast top.

(2) a 200 pound horizontal pull applied at a distance of 24 inches above the top of the mast.

(3) horizontal pulls of 60 pounds each applied at the midpoint of each mast section.

3.5.7.2 Guyed at launcher only .- The mast, guyed at the launcher only, shall be capable of meeting the requirement of 3.5.7 under the following conditions:

a. with each of the launcher guys pretensioned to 150 pounds.

b. with all other guys removed.

c. with the following forces all applied simultaneously on the same side of the mast and in the same plane:

(1) same as in 3.5.7.1b(1)

(2) a 24 pound pull applied as in 3.5.7.1b(2)

(3) 6 pound pulls applied as in 3.5.7.1b(3)

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3.5.8 Service conditions.- The mast, with all accessories, shall be capable of being subjected to the exposure conditions of 3.5.8.1, 3.5.8.2, 3.5.8.3 and 3.5.8.4 with no corrosion, winch lubricant leakage, pitting, blistering or loosening of finish or metal surfaces, hardening, cracking, softening or tackiness of rubber or plastic materials, decomposition of cloth, bonding failure, deformation, or other deterioration of materials. In addition, the mast shall be capable of the following operations with no malfunction or deterioration in ease of operation when performed at the steps specified in 3.5.8.1, 3.5.8.2, 3.5.8.3 and 3.5.8.4:

a. Operation of locking pins and locking screws.

b. Mounting of winch on launcher.

c. Operation of elevator with mast section in place.

d. Operation of brake with mast section in place.

e. Installation of guy plates and mast cap.

f. Rotation of azimuth (with mast section) and each guy plate independently.

g. Installation of mast sections in mast carrier and locking of handle.

h. Installation of mast carrier in launcher and locking in place.

j. Operation of the turnbuckle nut, the clamp handle and the lock on a tensioner and the pulling of a length of cable thru the tensioner.

3.5.8.1 High temperature exposure.- The mast shall be capable of meeting the requirements of 3.5.8 when subjected to Method 501, Procuedure I of MIL-STD-810 and the following conditions:

a. with Step 3 maintained for not less than 24 hours.

b. with +150°F temperature at Step 4.

c. with the operations of 3.5.8 performed at Steps 5 and 7.

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3.5.8.2 Low temperature exposure.- The mast shall be capable of meeting the requirements of 3.5.8 when subjected to Method 502, Procedure I of MIL-STD-810 and the following conditions:

a. with Step 2 maintained for not less than 24 hours.

b. with -80°F temperature at Step 2 and -65°F temperature at Step 4.

c. with the operations of 3.5.8 performed at Steps 5 and 7.

d. after meeting the requirements of 3.5.8.1.

3.5.8.3 Humidity exposure.- The mast shall be capable of meeting the requirements of 3.5.8 when subjected to Method 507, Procedure II of MIL-STD-810 and the following conditions:

a. with the operations of 3.5.8 performed at Steps 4 and 8 and during the 24th to 28th hours of each cycle of Step 6.

b. after meeting the requirements of 3.5.8.2.

3.5.8.4 Salt atmosphere exposure.- The mast shall be capable of meeting the requirements of 3.5.8 when subjected to Method 509 of MIL-STD-810 and the following conditions:

a. with those items, normally stored in bags, removed and exposed.

b. with the operations of 3.5.8 performed after exposure and after drying.

c. after meeting the requirements of 3.5.8.3.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any

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other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections.- The inspection requirements specified herein are classified as follows:

a. First article inspection (Does not include preparation for delivery). (See 4.3)

b. Inspections covered by subsidiary documents. (See 4.4)

c. Quality conformance inspections.

(1) Quality conformance inspection of equipment before preparation for delivery. (See 4.5)

(2) Quality conformance inspection of preparation for delivery. (See 4.19)

4.3 First article.- Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor. (See 3.2)

4.3.1 First article units.- The contractor shall furnish two (2) first article units of the complete Mast AB-577( )/GRC.

4.3.2 First article inspection.- The first article inspection shall consist of inspections specified in Table I, and shall be performed in the order specified in 3.2.

4.3.3 First article data.- The first article test plan and and test report shall be as required in the contract or purchase order.

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TABLE I - FIRST ARTICLE INSPECTION

Inspection	Req't Para	Insp Para
Subsidiary documents	3.3.9	4.4
Visual and mechanical	3.3.8	4.6
Brake locking and holding	3.5.1	4.7
Coupling closing	3.5.2	4.8
Interchangeability	3.3.10	4.9
Torsion transfer	3.5.3	4.10
Joint bending	3.5.4	4.11
Bounce, loose cargo	3.5.5	4.12
Shock, drop	3.5.6	4.13
Static load	3.5.7	4.14
Fungus	3.4.1.6	4.15
Temperature exposure	3.5.8.1 & 3.5.8.2	4.16
Humidity exposure	3.5.8.3	4.17
Salt atmosphere exposure	3.5.8.4	4.18

4.4 Inspection covered by subsidiary documents.- The equipment shall be inspected under the subsidiary documents specified in 3.3.9 as part of the inspection required by this specification, and the inspection requirement specified in the contract or purchase order.

4.5 Quality conformance inspection of equipment before preparation for delivery.- The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A inspection.- Each unit on contract or purchase order shall be inspected for conformance to the inspections specified in Table II. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection, utilizing the procedures of MIL-STD-105, using the general inspection levels, and AQL's indicated in Table II.

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4.5.1.1 Order of inspection within Group A.- Group A inspection shall be performed in an order satisfactory to the Government, except that visual and mechanical inspection shall be first.

TABLE II - GROUP A INSPECTION

Inspection	Req't Para	Insp Para	AQL Major	AQL Minor
Visual and mechanical	3.3.8	4.6	1.5 dphu	6.5 dphu
Brake locking and holding	3.5.1	4.7	1%	*
Coupling closing	3.5.2	4.8	1%	*

* All defects are considered major

4.5.2 Group B inspection.- Group B inspection shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection. This inspection shall conform to Table III and to the special inspection levels of Table I of MIL-STD-105.

4.5.2.1 Group B sampling plan.- The Group B AQL shall be 6.5 percent defective and the inspection level shall be S-4.

TABLE III - GROUP B INSPECTION

Inspection	Req't Para	Insp Para
Interchangeability	3.3.10	4.9

4.5.3 Group C inspection.- Group C inspection shall be performed on units that have passed Group A and B inspections. The inspection shall consist of the inspections specified in Table IV. Samples shall be selected in accordance with 4.5.3.1.

4.5.3.1 Sampling for Group C inspection.- Sample units shall be selected at random for the Group C inspection listed in Table IV in accordance with the following:

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4.5.3.1.1 Subgroup I.- One unit from each 25 production units or fraction thereof shall be subjected to the inspections of subgroup I in the order given in Table IV.

4.5.3.1.2 Subgroup II.- One unit from each 200 production units or fraction thereof shall be subjected to the inspections of subgroup II in the order given in Table IV.

4.5.3.1.3 Subgroup III.- Two units from the first 40 production units or fraction thereof shall be subjected to the inspections of subgroup III as follows:

a. One unit shall be subjected to the bounce and fungus inspections in that order.

b. The second unit shall be subjected to the temperature, humidity and salt fog inspections in that order.

TABLE IV - GROUP C INSPECTION

Inspection	Req't Para	Insp Para
Subgroup I		
Torsion transfer	3.5.3	4.10
Joint bending	3.5.4	4.11
Subgroup II		
Shock, drop	3.5.6	4.13
Static load	3.5.7	4.14
Subgroup III		
Bounce, loose cargo	3.5.5	4.12
Fungus	3.4.1.6	4.15
Temperature exposure	3.5.8.1 & 3.5.8.2	4.16
Humidity exposure	3.5.8.3	4.17
Salt atmosphere exposure	3.5.8.4	4.18

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4.5.3.2 Group C failures.- Action required relative to Group C failures shall be as specified in the contract or purchase order.

4.5.4 Reinspection of conforming Group C sample units.- Unless otherwise specified sample units which have been subjected to and passed Group C inspection may be accepted on the contract provided all damage is repaired and the sample units are resubjected to and pass Group B and Group A inspection in that order.

4.6 Visual and mechanical inspection.- The equipment shall be examined for the requirements of 3.3.8 and the defects listed in MIL-STD-252.

4.7 Brake locking and holding test.- The brake shall be tested under the conditions of, and meet the requirements of 3.5.1.

4.8 Coupling closing test.- The couplings shall be tested under the conditions of, and meet the requirements of 3.5.2.

4.9 Interchangeability inspection.- The equipment shall be gaged or measured per, and meet the requirements of 3.3.10. A dimension not within specified or design limits shall be considered a major defect.

4.10 Torsion transfer test.- The mast section assembly shall be tested under the conditions of, and meet the requirements of 3.5.3.

4.11 Joint bending test.- The mast sections shall be tested under the conditions of, and meet the requirements of 3.5.4.

4.12 Bounce, loose cargo test.- The mast shall be tested under the conditions of, and meet the requirements of 3.5.5.

4.13 Shock, drop test.- The mast shall be tested under the conditions of, and meet the requirements of 3.5.6.

4.14 Static load tests.- The mast shall be tested under the conditions of 3.5.7.1 and 3.5.7.2 and meet the requirements of 3.5.7.

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4.15 Fungus test.- The mast shall be tested under the conditions of, and meet the requirements of 3.4.1.6.

4.16 Temperature exposure test.- The mast shall be tested under the conditions of 3.5.8.1 and 3.5.8.2 and meet the requirements of 3.5.8.

4.17 Humidity exposure test.- The mast shall be tested under the conditions of 3.5.8.3 and meet the requirements of 3.5.8.

4.18 Salt atmosphere exposure test.- The mast shall be tested under the conditions of 3.5.8.4 and meet the requirements of 3.5.8.

4.19 Quality conformance inspection of preparation for delivery.- Preparation for delivery shall be inspected in accordance with MIL-P-116 to determine conformance to the requirements of Section 5.

4.20 Rough handling test (preparation for delivery).- When the rough handling test in accordance with MIL-P-116 is required by the contract (See 6.2c), the operations of 3.5.8 shall be performed to determine freedom from operational malfunction caused by rough handling.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.- Preservation and packaging shall be level A or C, as specified. (See 6.2b)

5.1.1 Level A.-

5.1.1.1 Cleaning.- Mast AB-577( )/GRC shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying.- Mast AB-577( )/GRC shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservation application.- None required.

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5.1.1.4 Unit packaging.- Unit packaging shall be in accordance with the methods prescribed in MIL-P-116 as specified herein.

5.1.1.4.1 Technical literature and guy bag with contents.- Each technical literature and bag with contents shall be packaged in accordance with ITEMS 1 and 2 of TABLE I and FIGURE 1.

5.1.1.4.2 Stake roll and contents.- Each stake roll and contents shall be packaged in accordance with ITEM 1 of TABLE II and FIGURE 2.

5.1.1.4.3 Mast AB-577( )/GRC.- Each mast shall be packaged in accordance with ITEMS 1 through 6 of TABLE III and FIGURE 3.

5.1.2 Level C.- Mast AB-577( )/GRC shall be preserved and packaged in a manner that will afford adequate protection against physical and environmental damage during shipment, handling and limited in transit storage.

5.2 Packing.- Packing shall be level A, B or C as specified. Shipping containers for all levels shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents. (See 6.2b)

5.2.1 Level A.- Units of Mast AB-577( )/GRC shall be packed as follows:

5.2.1.1 Technical literature and guy bag with contents.- Each technical literature and bag with contents shall be packed in accordance with ITEM 3 of TABLE I and FIGURE 1.

5.2.1.2 Stake roll with contents.- Each stake roll with contents shall be packed in accordance with ITEM 2 of TABLE II and FIGURE 2.

5.2.1.3 Mast AB-577( )/GRC.- Each mast shall be packed in accordance with ITEMS 7 and 8 of TABLE III and FIGURE 3.

5.2.2 Level B.- Units of Mast AB-577( )/GRC shall be packed as specified in 5.2.1.

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BILL OF MATERIAL
TABLE .
Box 1 of 3

REF No.	ITEM Nomenclature	No. Reqd	Method MIL-P-116	SIZE - INCHES L	W	D	REQUIREMENT Spec No.	T	C	V	G	S
1	Technical Literature	1	IC-1	10-1/2	8	1/4						
	Bag	1		12	9-1/2		MIL-B-117	II	C			
2	Guy Bag and Contents	1	III	22	14	15	(Wt. of bag and contents 64 lbs.)					
3	Wirebound Box	1		24-1/2	14-1/2	15-1/2	PPP-B-585		3			

T - Type C - Class V - Variety G - Grade S - Style

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BILL OF MATERIAL
TABLE II
Box 2 of 3

REF No.	ITEM Nomenclature	No. Reqd	Method MIL-P-116	SIZE - INCHES L	W	D	REQUIREMENT Spec No.	T	C	V	G	S
1	Stake Roll SM-C-423619 and contents	1	III	38	12-3/8	5-3/8	(Wt. of bag and contents 50 lbs.)					
2	Wirebound Box	1		40-1/2	13	6	PPP-B-585		3			3

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BILL OF MATERIAL
TABLE III
Box 3 of 3

REF No.	ITEM Nomenclature	No. Reqd	Method MIL-P-116	SIZE - INCHES L	W	D	REQUIREMENT Spec No.	T	C	V	G	S
1	Mast AB-577( )/GR	1	III	95-1/2	15-13/16	26	(Base item wt.	77 lbs.)				
2	Block, Wood	1		14-1/8	4-3/8	1-13/16	MIL-STD-731					
3	Block, Wood	1		14-1/8	2-3/8	1-13/16	MIL-STD-731					
4	Block, Wood	1		14-1/8	1-3/4	1-13/16	MIL-STD-731					
5	Nail, Screw Drive	12		2-1/2								
6	Corrugated Padding	7		62-3/4	1-3/4	5/32	PPP-F-320	CF	WR	SW	W5c	
7	Crate, Wood, Wirebound	1		97-7/8	16-9/16	28-9/16	MIL-C-11133	II	2		A	
8	Strapping, Metal	2		138	1/2	0.020	QQ-S-781	I	B		2	

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GUY BAG AND CONTENTS

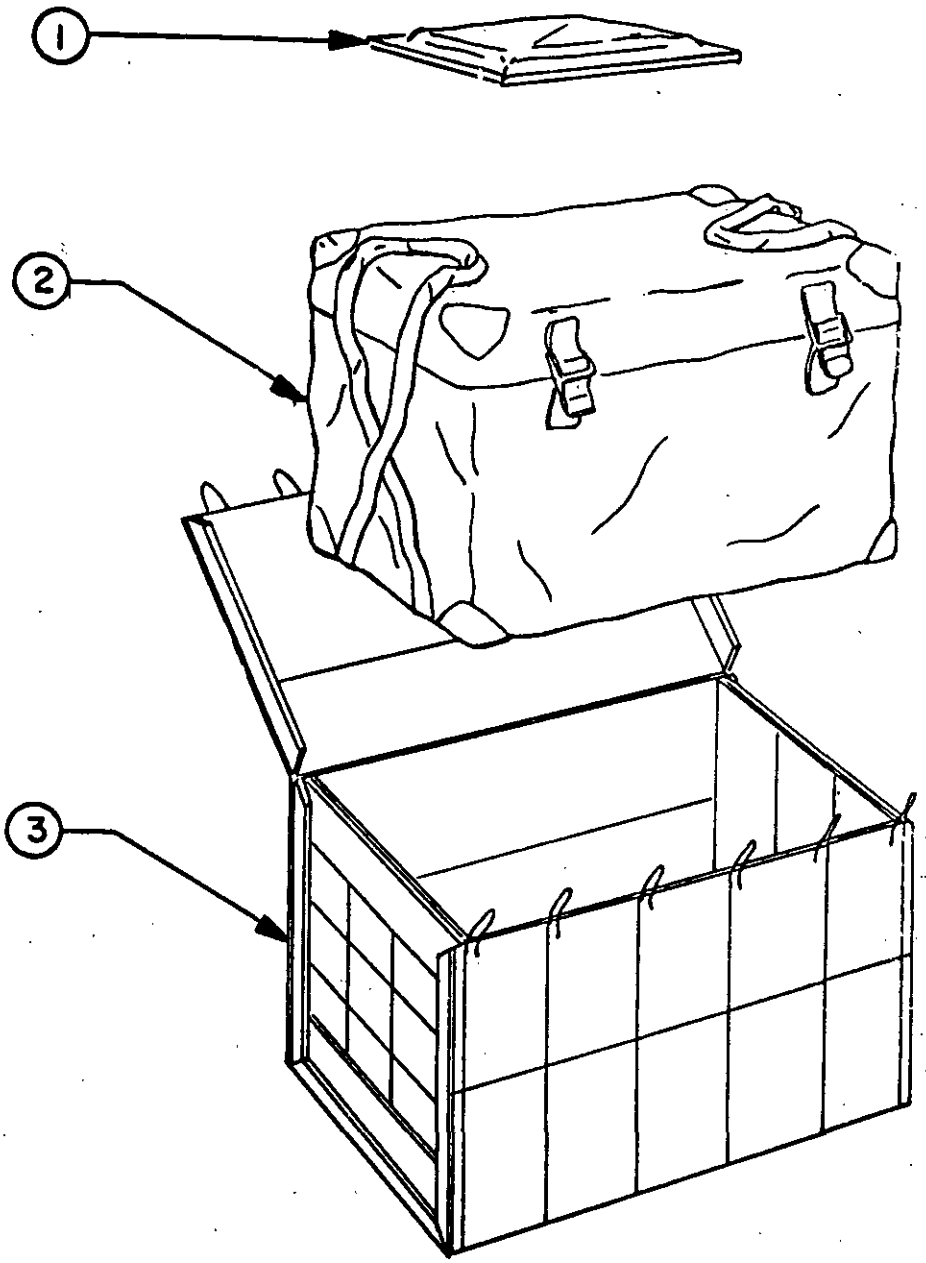


FIGURE 1

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STAKE ROLL AND CONTENTS

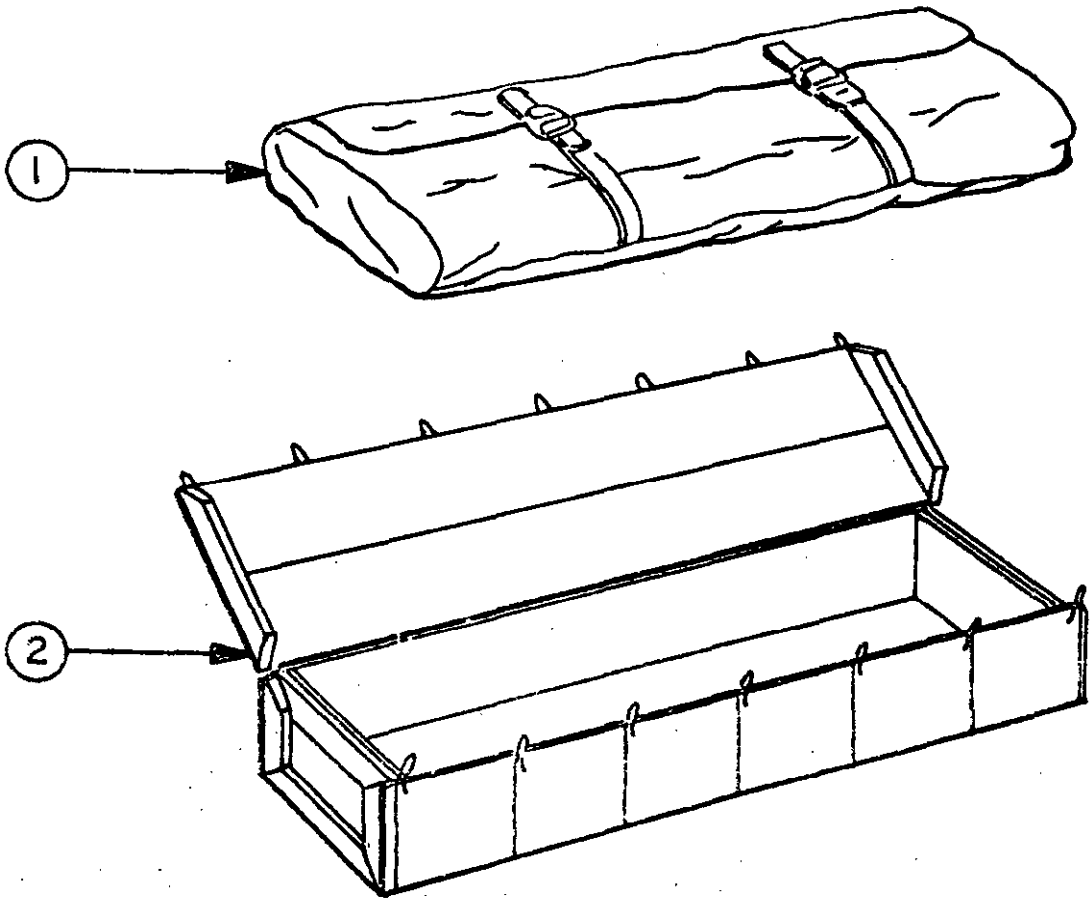


FIGURE 2

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MAST AB-577( )/GRC

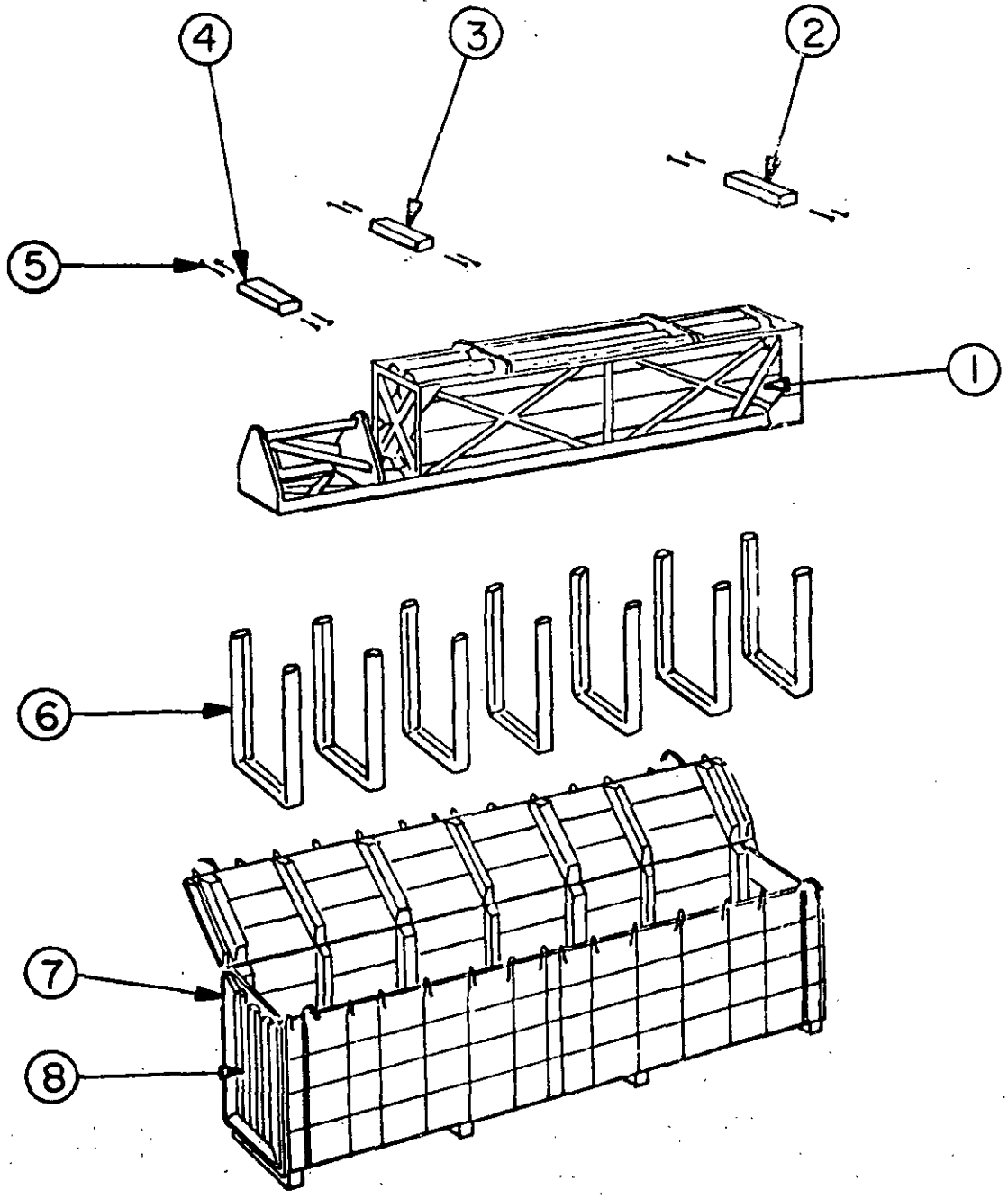


FIGURE 3

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5.2.3 Level C.- Units of Mast AB-577( )/GRC, packaged as specified in 5.1, shall be packed in shipping containers that comply with Federal or Military specifications, in a manner that will afford adequate protection to the package and its contents against physical and environmental deterioration and damage during limited shipment, handling and in transit storage.

5.3 Marking.- In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.- Mast AB-577( )/GRC is used for support of Radio Antennas such as AT-903( )/G and other general purpose equipment.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number, and date of this specification and any amendment thereto.

b. Levels of preservation and packaging and packing (See 5.)

c. When rough handling and functional tests are required. (See 4.20)

d. First article inspection: Two sample units of Mast AB-577( )/GRC are generally required so that lengthy environmental tests can be completed on one sample unit while complete performance measurements can be made on the second sample unit (See 4.3)

e. Marking and shipping of samples.

f. Place of final inspection.

g. Technical literature required.

h. Quantity of tools and running spare parts required.

i. Submission of the statement of treatment referenced in 3.11.5, as soon as possible after award of contract. This statement should be submitted to the contracting officer.

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6.3 Deviations.- Request for deviations in packaging and packing design shall be supported by test data proving the ability of the equipment, packaged and packed in accordance with the proposed design change to withstand the applicable tests for the selected method, including the rough handling tests in accordance with MIL-P-116. Deviation approval shall be contingent on availability of packaging and packing materials selected and economies realized. (See 5.)

6.4 Dimensional data.- Dimensions of packaging materials in the Bill of Material Table are based on the dimensions of the equipment cited in the tables. When dimensions of the equipment vary from those cited, the dimensions of the packaging materials shall be adjusted accordingly.

6.5 Nomenclature.- The parenthesis in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: AB-577/GRC or AB-577A/GRC. The contractor should apply for nomenclature in accordance with the applicable clause in the contract. (See 1.1)

6.6 Verification inspection.- Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

Custodian:
Army-EL

Preparing Activity
Army-EL

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U.S.Army Electronics Command
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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-M-55220A(EL) MAST AB-577()/GRC

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FM 1068-68

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